

Calculations for network paper

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PRELIMINARIES

These are preliminaries identities that can be derived by using the finite series identities in PRX. I have derived the following for $|m| < N$ and $m \neq 0$

$$\frac{1}{N} \sum_{k=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{1 - \cos \left[\frac{\pi k}{N} \right]} = 1 \quad (1)$$

Another identity exploits the fact that

$$\begin{aligned} [1 - (-1)^k] \sin \left[\frac{|m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right] &= \frac{1}{2} \left\{ \cos \left[\frac{(|m| - 1)\pi k}{N} \right] - \cos \left[\frac{(|m| + 1)\pi k}{N} \right] \right. \\ &\quad \left. - \frac{1}{2} \left[\cos \left[\frac{(N - |m| + 1)\pi k}{N} \right] + \cos \left[\frac{(N + |m| - 1)\pi k}{N} \right] - \cos \left[\frac{(N - |m| - 1)\pi k}{N} \right] - \cos \left[\frac{(N + |m| + 1)\pi k}{N} \right] \right] \right\} \end{aligned}$$

for a generic σ we have

$$\begin{aligned} \sum_{k=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{\sigma - \cos \left[\frac{\pi k}{N} \right]} &= \frac{1}{2} \sum_{k=1}^N \frac{\cos \left[\frac{(|m| - 1)\pi k}{N} \right] - \cos \left[\frac{(|m| + 1)\pi k}{N} \right]}{\sigma - \cos \left[\frac{\pi k}{N} \right]} \\ &\quad - \frac{1}{4} \sum_{k=1}^{N-1} \frac{\cos \left[\frac{(N - |m| + 1)\pi k}{N} \right] + \cos \left[\frac{(N + |m| - 1)\pi k}{N} \right] - \cos \left[\frac{(N - |m| - 1)\pi k}{N} \right] - \cos \left[\frac{(N + |m| + 1)\pi k}{N} \right]}{\sigma - \cos \left[\frac{\pi k}{N} \right]} \\ &= \frac{N}{2(\sigma^2 - 1)U_{N-1}(\sigma)} \left\{ T_{N-|m|+1}(\sigma) - T_{N-|m|-1}(\sigma) - T_{|m|-1}(\sigma) + T_{|m|+1}(\sigma) \right\} \end{aligned} \quad (2)$$

If $\sigma = \cos \left[\frac{2\pi r}{N} \right]$ one needs to use de l'Hospital rule to evaluate the expression above, which gives

$$\begin{aligned} \lim_{\sigma \rightarrow \cos \left[\frac{2\pi r}{N} \right]} \frac{N}{2(\sigma^2 - 1)U_{N-1}(\sigma)} \left\{ T_{N-|m|+1}(\sigma) - T_{N-|m|-1}(\sigma) - T_{|m|-1}(\sigma) + T_{|m|+1}(\sigma) \right\} \\ = \cos \left[\frac{2\pi r|m|}{N} \right], \end{aligned} \quad (3)$$

in other words we have the general identity for $0 < |m| < N$ and $r = 1, \dots, N - 1$,

$$\frac{1}{N} \sum_{k=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{\cos \left[\frac{2\pi r}{N} \right] - \cos \left[\frac{\pi k}{N} \right]} = \cos \left[\frac{2\pi r|m|}{N} \right], \quad (4)$$

The third relevant identity for $|m| \leq N$ and integer k is given by

$$\frac{1}{N} \sum_{r=1}^{N-1} \frac{\cos \left[\frac{2\pi r m}{N} \right]}{\cos \left[\frac{(2k-1)\pi}{N} \right] - \cos \left[\frac{2\pi r}{N} \right]} = \frac{1}{N \left\{ 1 - \cos \left[\frac{(2k-1)\pi}{N} \right] \right\}} - \frac{\sin \left[\frac{(2k-1)\pi|m|}{N} \right]}{\sin \left[\frac{(2k-1)\pi}{N} \right]} \quad (5)$$

alternatively

$$\frac{1}{N} \sum_{r=1}^{N-1} [1 - (-1)^k] \frac{\cos \left[\frac{2\pi r m}{N} \right]}{\cos \left[\frac{k\pi}{N} \right] - \cos \left[\frac{2\pi r}{N} \right]} = [1 - (-1)^k] \left\{ \frac{1}{N \left\{ 1 - \cos \left[\frac{\pi k}{N} \right] \right\}} - \frac{\sin \left[\frac{\pi k|m|}{N} \right]}{\sin \left[\frac{\pi k}{N} \right]} \right\} \quad (6)$$

PROPAGATOR IN RING LATTICE WITH ABSORPTION SITE

The occupation probability for the periodic ring lattice with arbitrary K and an absorption site at $n = m$ can be written via the general relation for any $t \geq 1$

$$W_{n_0}(n, t) = Q_{n_0}^{(k)}(n, t) - \sum_{s=1}^t \mathcal{F}_{n_0 \rightarrow m}(s) Q_m^{(k)}(n, t-s) \quad (7)$$

where $Q_r^{(k)}(n, t)$ is the propagator for the periodic ring lattice in the absence of any absorption site. Note that at $t = 0$, by construction $F_{n_0 \rightarrow m}(0) = 0$ when $n_0 \neq n$, and we have $W_{n_0}(n, 0) = Q_{n_0}^{(k)}(n, 0)$.

More explicit expressions can be obtained for the case $K = 2$, that is the nearest-neighbour lattice walk. We have that

$$Q_{n_0}^{(1)}(n, t) = \frac{1}{N} \sum_{k=0}^{N-1} \cos \left[\frac{2\pi k(n - n_0)}{N} \right] \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]^t, \quad (8)$$

and for $t \geq 1$

$$\mathcal{F}_{n_0}(m, t) = \frac{q}{N} \sum_{k=1}^{N-1} \left[1 - (-1)^k \right] \sin \left[\frac{|m - n_0|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right] \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^{t-1}. \quad (9)$$

with $\mathcal{F}_{n_0}(m, 0) = 0$.

Given that in Eq. (9) only odd k terms are non-zero, the first-passage probability can be written in compact form as

$$\mathcal{F}_{n_0}(m, t) = \sum_{k=1}^{\frac{N}{2}} f_k(n_0, m) \alpha_k^{t-1}. \quad (10)$$

where

$$f_k(n_0, m) = \frac{2q}{N} \sin \left[\frac{|m - n_0|\pi(2k-1)}{N} \right] \sin \left[\frac{\pi(2k-1)}{N} \right] \quad (11)$$

and

$$\alpha_k = 1 - q + q \cos \left[\frac{\pi(2k-1)}{N} \right]. \quad (12)$$

The generating function of Eq. (10) in compact form can be written as

$$\tilde{\mathcal{F}}_{n_0}(m, z) = z \sum_{k=1}^{N-1} \frac{f_k(n_0, m)}{1 - z\alpha_k} = \frac{U_{N-|m-n_0|-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) + U_{|m-n_0|-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right)}{U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right)}. \quad (13)$$

where $U_r(y)$ is the Chebyshev polynomial of the second kind.

Also we note that using Eq. (1) one can show that the survival probability

$$\sum_{s=1}^t \mathcal{F}_{n_0}(m, s) = 1 - \frac{1}{N} \sum_{k=1}^{N-1} \frac{\left[1 - (-1)^k \right] \sin \left[\frac{|n_0-m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{1 - \cos \left[\frac{\pi k}{N} \right]} \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t,$$

which provides the survival probability

$$R_{n_0}(t) = 1 - \sum_{s=0}^t \mathcal{F}_{n_0}(m, s) = \frac{1}{N} \sum_{k=1}^{N-1} \frac{\left[1 - (-1)^k \right] \sin \left[\frac{|n_0-m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{1 - \cos \left[\frac{\pi k}{N} \right]} \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t.$$

Using the above expressions in Eq. (7) and evaluating the time summation, we obtain

$$\begin{aligned}
W_{n_0}(n, t) &= \frac{1}{N} \sum_{k=1}^{N-1} \cos \left[\frac{2\pi k(n - n_0)}{N} \right] \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]^t \\
&+ \frac{1}{N^2} \sum_{k=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|n_0 - m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{1 - \cos \left[\frac{\pi k}{N} \right]} \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t \\
&+ \frac{1}{N^2} \sum_{k=1}^{N-1} \sum_{r=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|n_0 - m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right] \cos \left[\frac{2\pi r(n - m)}{N} \right]}{\cos \left[\frac{\pi k}{N} \right] - \cos \left[\frac{2\pi r}{N} \right]} \left\{ \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t - \left[1 - q + q \cos \left(\frac{2\pi r}{N} \right) \right]^t \right\}
\end{aligned} \tag{14}$$

Using the identities (4) and (6), we reduce the double summation to single summations as follows

$$\begin{aligned}
&\sum_{k=1}^{N-1} \sum_{r=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|n_0 - m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right] \cos \left[\frac{2\pi r(n - m)}{N} \right]}{\cos \left[\frac{\pi k}{N} \right] - \cos \left[\frac{2\pi r}{N} \right]} \left\{ \left[1 - q + q \cos \left(\frac{2\pi r}{N} \right) \right]^t - \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t \right\} \\
&= -N \sum_{r=1}^{N-1} \cos \left[\frac{2\pi r|n_0 - m|}{N} \right] \cos \left[\frac{2\pi r(n - m)}{N} \right] \left[1 - q + q \cos \left(\frac{2\pi r}{N} \right) \right]^t \\
&- \sum_{k=1}^{N-1} [1 - (-1)^k] \left\{ \frac{\sin \left[\frac{|n_0 - m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{1 - \cos \left[\frac{\pi k}{N} \right]} - N \sin \left[\frac{|n_0 - m|\pi k}{N} \right] \sin \left[\frac{n - m|\pi k}{N} \right] \right\} \\
&\times \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t \\
&= -N \sum_{r=1}^{N-1} \cos \left[\frac{2\pi r|n_0 - m|}{N} \right] \cos \left[\frac{2\pi r(n - m)}{N} \right] \left[1 - q + q \cos \left(\frac{2\pi r}{N} \right) \right]^t \\
&- \sum_{k=1}^{N-1} [1 - (-1)^k] \left\{ \frac{\sin \left[\frac{|n_0 - m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{1 - \cos \left[\frac{\pi k}{N} \right]} \right. \\
&\left. - N \sin \left[\frac{|n_0 - m|\pi k}{N} \right] \sin \left[\frac{n - m|\pi k}{N} \right] \right\} \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t
\end{aligned} \tag{15}$$

and we have

$$\begin{aligned}
W_{n_0}(n, t) &= \frac{1}{N} \sum_{k=1}^{N-1} \cos \left[\frac{2\pi k(n - n_0)}{N} \right] \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]^t \\
&- \frac{1}{N} \sum_{k=1}^{N-1} \cos \left[\frac{2\pi k(n - m)}{N} \right] \cos \left[\frac{2\pi k(n_0 - m)}{N} \right] \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]^t \\
&+ \frac{1}{N} \sum_{k=1}^{N-1} [1 - (-1)^k] \sin \left[\frac{\pi k|n_0 - m|}{N} \right] \sin \left[\frac{\pi k|n - m|}{N} \right] \left\{ 1 - q + q \cos \left[\frac{\pi k}{N} \right] \right\}^t.
\end{aligned} \tag{16}$$

As $\sum_{k=1}^{N-1} \cos \left[\frac{2\pi k(n - m)}{N} \right] \cos \left[\frac{2\pi k(n_0 - m)}{N} \right] = -1$ and $\sum_{k=1}^{N-1} [1 - (-1)^k] \sin \left[\frac{\pi k|n_0 - m|}{N} \right] \sin \left[\frac{\pi k|n - m|}{N} \right] = 0$, one can show that $Q_{n_0}(n, 0) = \delta_{n, n_0}$. Furthermore when either $n = m$ or $n_0 = m$, one has that $Q_{n_0}(n, t) = 0$ for all t , and it is also easy to see that $Q_{n_0}(n, t \rightarrow \infty) = 0$, which means that all random walks eventually reach the absorbing site $n = m$.

We can rewrite Eq. (16) as

$$W_{n_0}(n, t) = \sum_{k=1}^{N-1} G_k^{(1)}(n_0, n, m) \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]^t + G_k^{(2)}(n_0, n, m) \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t \tag{17}$$

where m is the absorbing site, and with

$$\begin{aligned} G_k^{(1)}(n_0, n, m) &= \frac{\cos \left[\frac{2\pi k(n-n_0)}{N} \right] - \cos \left[\frac{2\pi k(n-m)}{N} \right] \cos \left[\frac{2\pi k(n_0-m)}{N} \right]}{N} \\ G_k^{(2)}(n_0, n, m) &= \frac{\left[1 - (-1)^k \right] \sin \left[\frac{\pi k|n_0-m|}{N} \right] \sin \left[\frac{\pi k|n-m|}{N} \right]}{N} \end{aligned} \quad (18)$$

In terms of generating functions one has

$$\widetilde{W}_{n_0}(n, z) = \sum_{k=1}^{N-1} \frac{G_k^{(1)}(n_0, n, m)}{1 - z \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]} + \sum_{k=1}^{N-1} \frac{G_k^{(2)}(n_0, n, m)}{1 - z \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]} \quad (19)$$

or the equivalent form

$$\widetilde{W}_{n_0}(n, z) = \sum_{k=1}^{N-1} \frac{G_k^{(1)}(n_0, n, m)}{1 - z\gamma_k} + \sum_{k=1}^{\frac{N}{2}} \frac{g_k^{(2)}(n_0, n, m)}{1 - z\alpha_k}, \quad (20)$$

and its time-dependent version

$$W_{n_0}(n, t) = \sum_{k=1}^{N-1} G_k^{(1)}(n_0, n, m) \gamma_k^t + \sum_{k=1}^{\frac{N}{2}} g_k^{(2)}(n_0, n, m) \alpha_k^t, \quad (21)$$

where α_k is defined in Eq. (12),

$$\gamma_k = 1 - q + q \cos \left(\frac{2\pi k}{N} \right), \quad (22)$$

and

$$g_k^{(2)}(n_0, n, m) = \frac{2}{N} \sin \left[\frac{\pi(2k-1)|n_0-m|}{N} \right] \sin \left[\frac{\pi(2k-1)|n-m|}{N} \right]. \quad (23)$$

The associated expressions in terms of Chebyshev polynomials of the first and second kind, respectively, $T_r(y)$ and $U_r(y)$, are

$$\begin{aligned} \sum_{k=1}^{N-1} \frac{G_k^{(1)}(n_0, n, m)}{1 - z \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]} &= \left\{ 2T_{N-|n-n_0|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) + 2T_{|n-n_0|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right. \\ &\quad - T_{N-||n_0-m|-|n-m||} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) - T_{||n_0-m|-|n-m||} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \\ &\quad \left. - T_{N-[|n_0-m|+|n-m| \pmod{N}]} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) - T_{|n_0-m|+|n-m| \pmod{N}} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right\} \\ &\quad \times \left[2zq \left\{ \left[1 + \frac{1}{q} \left(\frac{1}{z} - 1 \right) \right]^2 - 1 \right\} U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right]^{-1}, \end{aligned} \quad (24)$$

and

$$\begin{aligned} \sum_{k=1}^{N-1} \frac{G_k^{(2)}(n_0, n, m)}{1 - z \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]} &= \left\{ T_{N-||n_0-m|-|n-m||} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) + T_{N-|N-|n_0-m|-|n-m||} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right. \\ &\quad \left. - T_{|N-|n_0-m|-|n-m||} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) - T_{||n_0-m|-|n-m||} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right\} \\ &\quad \times \left[2zq \left\{ \left[1 + \frac{1}{q} \left(\frac{1}{z} - 1 \right) \right]^2 - 1 \right\} U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right]^{-1}, \end{aligned} \quad (25)$$

When $n = n_0$ Eq. (17) reduces to

$$W_n(n, t) = \frac{1}{N} \sum_{k=1}^{N-1} \left\{ \left(1 - \cos^2 \left[\frac{2\pi k(n-m)}{N} \right] \right) \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]^t + \left[1 - (-1)^k \right] \sin^2 \left[\frac{\pi k|n-m|}{N} \right] \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t \right\}, \quad (26)$$

whose generating function is given by the compact expression

$$\begin{aligned} \widetilde{W}_n(n, z) = & \left\{ 2T_N \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) + T_{N-|N-2|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) - T_{|N-2|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right. \\ & \left. - T_{N-[2|n-m| \pmod{N}]} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) - T_{[2|n-m| \pmod{N}]} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right\} \\ & \left\{ 2zq \left\{ \left[1 + \frac{1}{q} \left(\frac{1}{z} - 1 \right) \right]^2 - 1 \right\} U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right\}^{-1} \end{aligned} \quad (27)$$

Note that using Eqs. (19) with $n = n_0$ and (27) when $|n-m| = N/2$ with N even, one can show setting $z = 1$ the following identities

$$\sum_{k=1}^{N-1} \frac{1 - (-1)^k}{1 - \cos \left(\frac{\pi k}{N} \right)} = \frac{N^2}{2}, \quad \sum_{k=1}^{N-1} \frac{1}{1 - \cos \left(\frac{\pi k}{N} \right)} = \frac{2N^2 + 1}{6}, \quad \sum_{k=1}^{N-1} \frac{(-1)^k}{1 - \cos \left(\frac{\pi k}{N} \right)} = \frac{1 - N^2}{6}.$$

For the special case $|n-m| = N/2$ with N even Eq. (27) becomes

$$\widetilde{W}_n(n, z) = \frac{T_N \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) - 1}{zq \left\{ \left[1 + \frac{1}{q} \left(\frac{1}{z} - 1 \right) \right]^2 - 1 \right\} U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right)} = \frac{2 \left[U_{\frac{N}{2}-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right]^2}{zq U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right)} \quad (28)$$

PROPAGATOR WITH ONE SHORTCUT AND ONE ABSORBING SITE

We start from the general propagator with a shortcut from $n = u$ to site $n = v$ and we have

$$\widetilde{S}_{n_0}(n, z) = \widetilde{P}_{n_0}(n, z) - \frac{z\beta(1-q)\widetilde{P}_{n_0}(u, z) \left[\widetilde{P}_u(n, z) - \widetilde{P}_v(n, z) \right]}{1 + z\beta(1-q) \left[\widetilde{P}_u(u, z) - \widetilde{P}_v(u, z) \right]}, \quad (29)$$

where $\widetilde{P}_r(n, z)$ is the ring propagator generating function without shortcuts. If the ring also has an absorbing site, the modularity of the defect technique allows to replace the (probability conserving) propagator $\widetilde{P}_r(n, z)$ with the propagator $\widetilde{W}_r(n, z)$ in Eq. (17)

$$\widetilde{S}_{n_0}(n, z) = \widetilde{W}_{n_0}(n, z) - \frac{z\beta(1-q)\widetilde{W}_{n_0}(u, z) \left[\widetilde{W}_u(n, z) - \widetilde{W}_v(n, z) \right]}{1 + z\beta(1-q) \left[\widetilde{W}_u(u, z) - \widetilde{W}_v(u, z) \right]}. \quad (30)$$

When the absorbing site coincides with the shortcut site, i.e. $m = v$, $\widetilde{W}_v(n, z) = 0$ and Eq. (30) simplifies further to

$$\widetilde{S}_{n_0}(n, z) = \widetilde{W}_{n_0}(n, z) - \frac{z\beta(1-q)\widetilde{W}_{n_0}(u, z)\widetilde{W}_u(n, z)}{1 + z\beta(1-q)\widetilde{W}_u(u, z)}. \quad (31)$$

After summing Eq. (30) over n to obtain the survival probability $\tilde{\mathcal{S}}_{n_0}(z) = [1 - \tilde{F}_{n_0}(v, z)] / (1 - z)$, one expresses the generating function of the first-passage probability to site v as

$$\tilde{F}_{n_0}(v, z) = \tilde{\mathcal{F}}_{n_0}(v, z) + \frac{z\beta(1-q)\tilde{W}_{n_0}(u, z) [1 - \tilde{\mathcal{F}}_u(v, z)]}{1 - z\beta(1-q)\tilde{W}_u(u, z)}, \quad (32)$$

or more explicitly as

$$\tilde{F}_{n_0}(v, z) = \tilde{\mathcal{F}}_{n_0}(v, z) - qz\tilde{W}_{n_0}(u, z) [1 - \tilde{\mathcal{F}}_u(v, z)] \tilde{H}_{N,|u-v|}(z), \quad (33)$$

with $\tilde{W}_{n_0}(u, z)$ and $\tilde{\mathcal{F}}_r(m, z)$ defined, respectively, in Eqs. (19) and (13), and

$$\tilde{H}_{N,|u-v|}(z) = \frac{\beta(1-q)}{q} \frac{1}{1 + z\beta(1-q)\tilde{W}_u(u, z)}. \quad (34)$$

We now consider the symmetric case $|u - v| = N/2$ with N even, and use $\tilde{W}_u(u, z)$ from Eq. (28) in $\tilde{W}_u(u, z)$, that allows to simplify Eq. (34) to

$$\tilde{H}_{N,N/2}(z) = \frac{1}{\frac{q}{\beta(1-q)} + \phi(z)}, \quad (35)$$

where

$$\phi(z) = \frac{2U_{N/2-1}^2 \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1\right]\right)}{U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1\right]\right)}. \quad (36)$$

whose time-dependence is $H_{N,N/2}(0) = \beta(1-q)/q$ and for $t \geq 1$ is given by the complex integral (after the change of integration variable $z = 1/s$ and the region of integration becoming $|s| \leq 1$)

$$H_{N,N/2}(t) = \frac{1}{2\pi i} \oint_{|s| \leq 1} \frac{s^{t-1}}{\frac{q}{\beta(1-q)} + \phi\left(\frac{1}{s}\right)}. \quad (37)$$

The denominator in the integrand of Eq. (37) has M real zeros s_k ($k = 1, 2, \dots, M$) and are shown graphically in Fig. 1 (see the figure caption for an explanation of when $M = N/2$ and $M = N/2 - 1$). s_k (real) poles in the integrand with $k = 1, \dots, N/2 - 1$ in the dominator ($-1 < s_k < 1$) can be found numerically as

$$H_{N,N/2}(t) = \sum_{k=1}^M c_k s_k^{t-1} \quad (38)$$

with

$$c_k = -\frac{N \left(1 + \frac{1}{N}\right) \left(1 + \frac{1}{q} [s_k - 1]\right) U_{N-1} \left(1 + \frac{1}{q} [s_k - 1]\right) - T_N \left(1 + \frac{1}{q} [s_k - 1]\right) - U_{N-2} \left(1 + \frac{1}{q} [s_k - 1]\right) - 1}{\left\{\left(1 + \frac{1}{q} [s_k - 1]\right)^2 - 1\right\} \left\{T_{\frac{N}{2}} \left(1 + \frac{1}{q} [s_k - 1]\right)\right\}^2}. \quad (39)$$

Analysis of c_k shows that $c_k < 0$ for any real s_k although one may have $c_k = -\infty$ when $s_k = \alpha_k$. But since α_k are the asymptotes of $\phi(1/s)$ and $s_k \neq \alpha_k$, then $c_k < 0$ and finite for all k .

The time dependence of Eq. (33) for $|u - v| = N/2$ is thus given by the time-convoluted expression for $t \geq 1$

$$F_{n_0}(v, t) = \mathcal{F}_{n_0}(v, t) - q \sum_{t'=0}^{t-1} W_{n_0}(u, t-1-t') \sum_{t''=0}^{t'} [\delta_{t',t''} - \mathcal{F}_u(v, t' - t'')] H_{N,N/2}(t'') \quad (40)$$

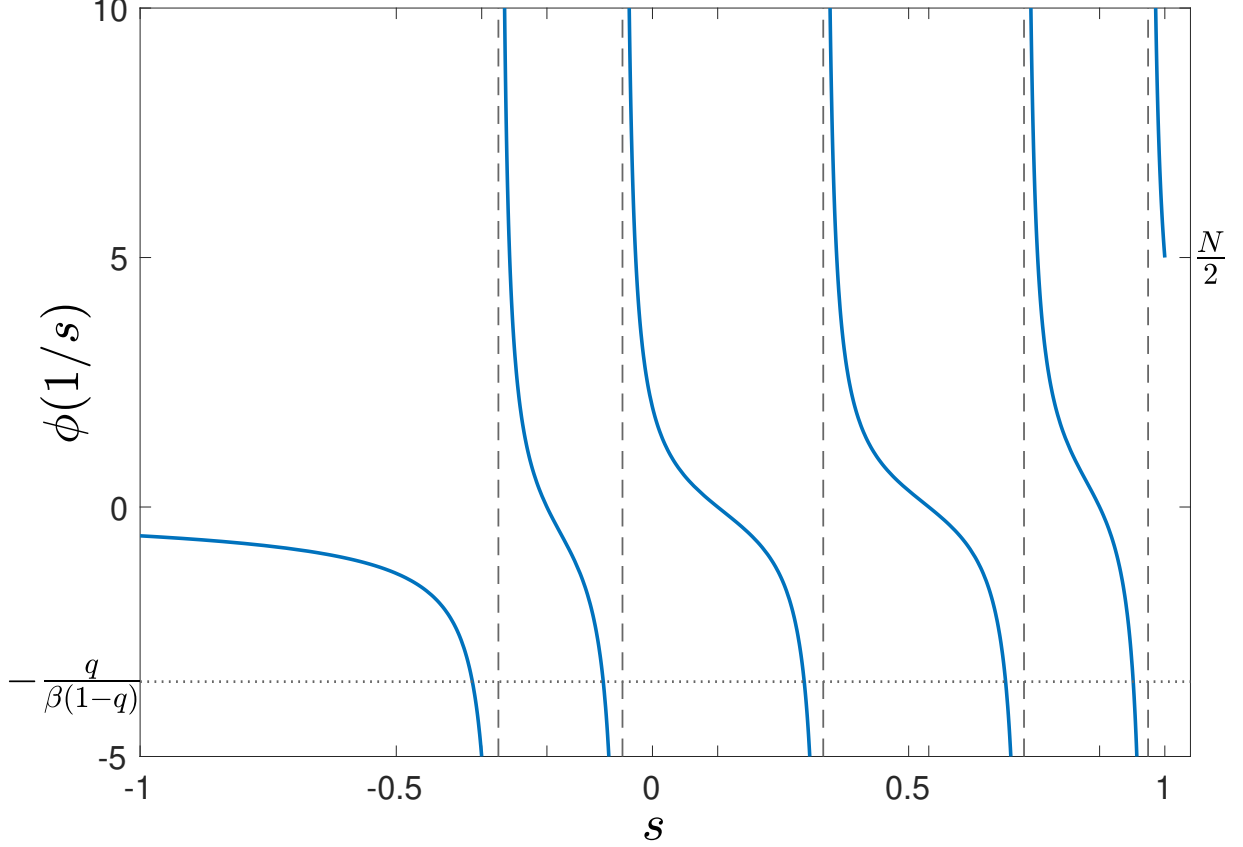


FIG. 1. Graphical representation of the poles of Eq. (35) for the case $N = 10$, $q = 2/3$ and $\beta = 4/7$ which are the values of s in the range $-1 \leq s \leq 1$ where the horizontal dotted line $-\frac{q}{\beta(1-q)}$ intercepts the function $\phi(s^{-1})$. Irrespective of N , β and q (N even, $0 < \beta \leq 1$ and $0 < q < 1$), there are $M = \frac{N}{2}$ poles at s_k . The asymptotes of the $\phi(s^{-1})$ function are given from right to left by, respectively, α_k in Eq. (12) with $k = 1, 2, \dots, \frac{N}{2}$, and they are interlaced with the poles s_k with $k = 1, \dots, M$. We follow the same principle and order from right to left the poles s_k with increasing k , thus with s_1 representing the largest positive pole. With $q = 1$ the curves are symmetric around the origin, but given the requirement $q > 1$ to have a non-zero probability to move from site u to site v , the function $\phi(s^{-1})$ is never symmetric around $s = 0$, which implies that in magnitude one has $|s_{\frac{N}{2}}| < s_1$ for any $q < 1$. The paired tick marks without numerical values on the top and bottom horizontal axes correspond, from right to left, to the values γ_k in Eq. (22) with $k = 1, 2, \dots, \frac{N}{2}$, which are interlaced with the α_k (the values for $N/2 + 1 \leq k \leq N - 1$ repeat themselves but we do not show the corresponding tick marks).

and $F_{n_0}(v, 0) = 0$. Using Eqs. (10), (21) and (38), after performing the various time convolutions, and assuming $s_k \neq \gamma_k$ one obtains

$$\begin{aligned}
 F_{n_0}(v, t) = & \sum_{\ell=1}^{\frac{N}{2}} f_{\ell}(n_0, u) \alpha_{\ell}^{t-1} - q \sum_{k=1}^M \left\{ \sum_{r=1}^{N-1} \frac{c_k G_r^{(1)}(n_0, u, v)}{s_k(s_k - \gamma_r)} (s_k^t - \gamma_r^t) + \sum_{\ell=1}^{\frac{N}{2}} \frac{c_k g_{\ell}^{(2)}(n_0, u, v)}{s_k(s_k - \alpha_{\ell})} (s_k^t - \alpha_{\ell}^t) \right\} \\
 & + q \sum_{k=1}^M \sum_{r=1}^{N-1} \sum_{\ell=1}^{\frac{N}{2}} \frac{c_k G_r^{(1)}(n_0, u, v) f_{\ell}(u, v)}{s_k - \alpha_{\ell}} \left[\frac{s_k^t - \gamma_r^t}{\alpha_{\ell}(s_k - \gamma_r)} - \frac{\alpha_{\ell}^t - \gamma_r^t}{s_k(\alpha_{\ell} - \gamma_r)} \right] - q \sum_{k=1}^M \sum_{\ell=1}^{\frac{N}{2}} \frac{c_k g_{\ell}^{(2)}(n_0, u, v) f_{\ell}(u, v)}{s_k \alpha_{\ell}(s_k - \alpha_{\ell})} \alpha_{\ell}^t t \\
 & + q \sum_{k=1}^M \underbrace{\sum_{r=1}^{\frac{N}{2}} \sum_{\ell=1}^{\frac{N}{2}}}_{\ell \neq r} \frac{c_k g_r^{(2)}(n_0, u, v) f_{\ell}(u, v)}{s_k \alpha_{\ell}(s_k - \alpha_{\ell})} \left[s_k \frac{s_k^t - \gamma_r^t}{s_k - \gamma_r} - \alpha_{\ell} \frac{\alpha_{\ell}^t - \gamma_r^t}{\alpha_{\ell} - \gamma_r} \right]. \tag{41}
 \end{aligned}$$

The exponentially decaying dominant term in the first-passage probability can now be evinced by finding the largest positive coefficient among α_k , γ_k and s_k with $k = 1$ identifying the largest value for each one. As $\alpha_1 > s_1$ and $\alpha_1 > \gamma_1$, the dominant term is the long time dependence dominated by $\alpha_1^t t$, which indicates a logarithmic correction that slows the exponential decay. In other words the introduction of the shortcut between u and v changes the long time dependence of the first-passage probability from α_1^t to $\alpha_1^t t$. As the second largest decaying exponential term is either γ_1 or s_1 , there is scope to explore whether the transition from having two peaks to one peak occurs when the second largest decaying exponential term switch from s_1^t to γ_1^t , which graphically one can see happens as β is sufficiently large. When that happens, i.e. when $s_1 = \gamma_1$, there will be corrections to the expression in Eq. (41) with terms of the form $\gamma_1^t t$ appearing.